CT-23025

UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

TSCS, TSCS(CYBER), TSCS(GAMING) Midterm Examinations Fall 2025

Time: 90 minutes | **Design and Analysis Of Algorithm (CT-363)** | **Max Marks: 20**

Note: Attempt all questions. | **Paper – A** | **Date: 07-10-2025**

Q1) 4 marks CL0 2

You are a freelance consultant with 5 potential projects, each with a profit (in dollars) and a deadline (in days from now). Projects take 1 day each to complete. Projects: A (profit 100, deadline 2), B (80, 1), C (120, 3), D (90, 2), E (110, 4). Use the greedy algorithm to schedule projects to maximize total profit without missing deadlines. What is the optimal schedule and total profit? Explain how this applies to prioritizing client work in a real consulting business.

Q2) 4 marks CL0 2

In photo editing software, metadata tags appear with frequencies: 'Exposure' (20), 'ISO' (10), 'Aperture' (25), 'Shutter' (15), 'Lens' (18), 'Filter' (12). Apply Huffman algorithm to generate codes. Compute the tree, codes, and compression ratio for a dataset with 200 tags (vs. 8-bit fixed). Discuss its role in optimizing storage for large photo libraries on devices. **Also design and algorithm for this.**

Q3) 4 marks CL0 2

An IoT system collects sorted timestamped sensor readings (e.g., temperature data) from a smart city's environmental monitoring network, stored in a sorted array on a low-memory device. The system needs to find a specific reading by timestamp for anomaly detection. Jump Search is used to balance efficiency and simplicity on resource-constrained devices. The system processes n readings. **Also design and algorithm for this.**

```
readings = [{"timestamp": 16200, "temperature": 24.0}, {"timestamp": 16300, "temperature": 24.2},{"timestamp": 16400, "temperature": 25.0},{"timestamp": 16500, "temperature": 23.8},{"timestamp": 16600, "temperature": 23.5},{"timestamp": 16700, "temperature": 24.1},{"timestamp": 16800, "temperature": 24.5},{"timestamp": 16900, "temperature": 24.8},{"timestamp": 17000, "temperature": 25.2}, {"timestamp": 17100, "temperature": 24.9}]target = 16800;
```

Q4) (2+6) marks CL0 1

a) Define designing techniques used in making an algorithms

b) Find time function, time complexity of recursive as well as non-recursive algorithms

1) let n = 5

```
void selectionSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        int min_idx = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[min_idx]) {
                min_idx = j;
            }
        }
        if (min_idx != i) {
            std::swap(arr[i], arr[min_idx]);
        }
    }
}
```

2) let n = 5

```
int i, j, k = 0;
for (i = n / 2; i <= n;i++)
{
for (j = 2; j <= n; j = j *
2) {
        k = k + n / 2;
    }
}
```

3) let n = 5

```
for (i = 0; i <= n; i++)
{
    cout<< i
}
Func(n)
{
If(n > 1){
for (i = 0; i <= n; i++)
{
    cout<< i
}
Func(n-1)
}
```

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Note: Attempt all questions. | **Paper – B** | **Date: 07-10-2025**

Q1)

4 marks CL0 2

A user has 4 backup files to merge into one archive: sizes 5GB, 8GB, 12GB, 7GB. Apply greedy optimal merge to calculate the sequence and total merge cost. Explain the cost savings compared to naive merging, and its application in cloud services like Google Drive for efficient storage management. **Also design and algorithm for this.**

Q2)

4 marks CL0 2

A router allocates 1000 Mbps bandwidth to 5 data streams: Stream1 (value 500 priority, weight 200 Mbps), Stream2 (400, 150), Stream3 (600, 250), Stream4 (550, 180), Stream5 (450, 220). Greedy fractional to maximize priority. Provide allocation and total, and explain QoS in real networks.

Q3)

4 marks CL0 2

An e-commerce warehouse maintains a sorted list of product SKUs (Stock Keeping Units) for inventory lookups during order fulfillment. The list is stored on a server, and searches are frequent to locate items for picking. Jump Search is used to minimize CPU cycles and disk I/O for sorted SKU lists. The warehouse has n SKUs. **Also design and algorithm for this.**

```
products = [{"sku": 1001, "name": "Laptop"},{"sku": 1002, "name": "Phone"},{"sku": 1003, "name":
"Tablet"},{"sku": 1004, "name": "Headphones"},{"sku": 1005, "name": "Charger"},{"sku": 1006, "name":
"Mouse"},{"sku": 1007, "name": "Keyboard"},{"sku": 1008, "name": "Monitor"}, {"sku": 1009, "name":
"Speaker"}, {"sku": 1010, "name": "Webcam"}]target = 1003;
```

Q4)

(2+6) marks CL0 1

a) Discuss the designing techniques used in making an algorithms

b) Analyze time function, time complexity of recursive as well as non-recursive algorithms

1) let n = 5

```
void selectionSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        int min_idx = i;
        for (int j = i + 1; j < n; j++) {
            if (arr[j] < arr[min_idx]) {
                min_idx = j;
            }
        }
        if (min_idx != i) {
            std::swap(arr[i], arr[min_idx]);
        }
}
```

2) let n = 5

```
int a = 0, b = 0;
for (i = 0; i < N; i++) {
    a = a + rand();
}
for (j = 0; j < M; j++) {
    b = b + rand();
}
```

3) let n = 5

```
for (i = 0; i <= n; i++) {      n+1
for (j = 0; j <= n; j++) {      n+1
    cout<< i<<j
}
}
Func(n)
{
If(n > 1){
for (i = 0; i <= n; i++) {
    cout<< i
}
Func(n-1)
}
```